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Enhanced logistic map with infinite chaos and its applicability in lightweight and high-speed pseudo-random bit generation

Mir Nazish^{1*} , Munika Javid¹ and M. Tariq Banday¹

Abstract

Chaotic maps are employed in cryptography and secure communications due to their unpredictable and complex dynamics. However, existing chaotic maps, specifically the one-dimensional chaotic maps, often have limited chaotic control parameter ranges, which restricts their effectiveness and applicability in practical low-end applications. This paper proposes an enhanced Logistic map with an infinite chaotic control parameter range to address these limitations. The performance of the proposed map has been comprehensively evaluated using various chaos dynamical tests, including the bifurcation diagram, Lyapunov exponent, cobweb plots, 2D and 3D phase plots, approximate entropy, and sample entropy, time sensitivity analysis, and the 0–1 test. The results demonstrate that the improved Logistic map significantly outperforms its seed map across all evaluation metrics. Additionally, the enhanced Logistic map-based pseudorandom bit generator (PRBG) has been designed and evaluated for resource efficiency and security. The findings validate that the PRBG achieves significant implementation efficiency while also successfully qualifying the fifteen NIST tests, validating its statistical randomness. Thus, the proposed map and the PRBG position themselves as lightweight, highly secure solutions for safeguarding resource-limited smart IoT applications.

Keywords Chaotic system, One-dimensional chaotic maps, Logistic map, NIST, Bifurcation diagram, Lyapunov exponent

Introduction

Chaotic maps (Zia et al. 2022; Muthu and Murali 2021a; Nazish and Banday 2023) are non-linear mathematical functions known for their deterministic yet highly unpredictable and complex dynamics. They are characterized by their extreme sensitivity to initial conditions, with a minuscule change in the input state resulting in vastly divergent output states. Additionally, chaotic maps are ergodic, ensuring that trajectories eventually explore every part of their phase space. They also exhibit topological mixing, causing nearby points on the map's domain to eventually diverge widely in their range.

These properties make chaotic maps valuable tools for resolving complex real-world problems across various domains (Grassi 2021; Zolfaghari and Koshiba Jun. 2022; Naik and Singh 2022; Mammedov et al. 2022). In biology, they model population dynamics and the spread of diseases, providing valuable insights for ecosystem management and disease control. Additionally, chaotic maps help understand economic behaviors in financial modeling by simulating market fluctuations. In engineering, they are used to design secure and efficient control systems. Furthermore, in cryptography, they are used to generate secure and unpredictable pseudorandom bit sequences, which are essential for ensuring robust security and reliability in IoT and H-IoT applications (Trujillo-Toledo et al. 2023; Naik and Singh 2024; Yu et al. 2019).

Chaotic maps are classified into discrete and analog (Muthu and Murali 2021a; Nazish and Banday 2023),

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based on their mathematical formulations and the nature of their outputs. Discrete maps are governed by difference equations, producing discrete output states over time. Notable examples include Logistic (May 1976), Sine (Griffin 2013), Chebyshev (Liu et al. 2016), and Singer (Tubishat et al. 2022) maps. In contrast, analog maps are described by differential equations, resulting in continuous trajectories in phase space. Examples of analog maps include Lorenz (Lorenz 2017) and Rossler (Zgliczyński 1997) maps, which demonstrate chaotic dynamics through continuous time evolution. Also, based on dimensionality, chaotic maps can be one-dimensional (1D), two-dimensional (2D), or higher-dimensional (Zia et al. 2022; Muthu and Murali 2021a; Nazish and Banday 2023). 1D maps, such as Logistic and Tent maps, operate on a single variable and often exhibit simple chaotic behavior. Two-dimensional and higher-dimensional maps involve two or more variables, producing even more non-linear and intricate chaotic behavior. Henon (Benedicks and Carleson 1991) and Lorenz maps are well-known two-dimensional and higher-dimensional maps.

Though the higher dimensional maps exhibit greater chaotic complexity, they are often unsuitable for securing low-end applications due to their computational complexity and higher resource requirements. For such applications, discrete 1D maps offer better security solutions due to their ease of implementation and analysis. Their mathematical simplicity allows for efficient computation, making them ideal for real-time embedded systems with limited computational resources. However, 1D maps suffer from limited chaotic control parameter ranges, which hinder their practical applicability for high-security demanding applications (Hua et al. Feb. 2019; Dong et al. 2021; Sen Teh et al. 2020). These suffer from chaos degradation on digital systems with finite precision, where the intended chaotic behavior becomes less predictable as the output states cannot be uniformly distributed. Also, as these maps exhibit more straightforward and less intricate chaotic behavior, their trajectories become more susceptible to prediction. Additionally, these maps generate chaotic behavior only within specific parameter ranges. In some maps, the chaotic behavior is discontinuous, with abrupt changes or gaps even within the chaotic trajectories. As such, external factors such as noise or slight parameter variations can easily disrupt or destroy the chaotic properties of the map. This makes them highly susceptible to exhaustive key search attacks and, therefore, are not considered a reliable solution for securing practical applications. This paper addresses the above limitations associated with the Logistic map, one of the most widely used discrete 1D chaotic maps, by proposing its enhanced version characterized by an infinite chaotic control parameter range. Additionally, the

paper introduces a novel approach to Pseudo-Random Bit Generator (PRBG) design using the enhanced Logistic map, demonstrating its feasibility and effectiveness in generating statistically random and highly unpredictable bit sequences.

The paper makes the following key contributions:

1. Introduces a novel enhanced Logistic map with an infinite chaotic control parameter range.
2. Conducts a rigorous evaluation of the proposed map through a series of chaos dynamical tests, including bifurcation diagram, Lyapunov exponent, 2D and 3D phase plots, cobweb plots, 0–1 test, approximate and sample entropies, and time sensitivity analysis.
3. Proposes a lightweight and high-speed pseudo-random bit-generator (PRBG) utilizing the enhanced Logistic map.
4. Evaluates the resource efficiency and speed of the proposed PRBG.
5. Assesses the generated bit sequence against fifteen NIST tests for statistical randomness.

The rest of the paper is organized as follows: Section "Related Work" reviews research on enhancing existing 1D maps. Section "Proposed Enhanced Logistic Map" introduces the proposed enhanced Logistic map. Section "Proposed Pseudo-Random Bit Generator based on the Enhanced Logistic Map" describes the implementation methodology and presents the proposed map's evaluation results in various chaos dynamical tests. Section 5 discusses the proposed pseudo-random bit generator based on the enhanced Logistic map and presents its performance and security analysis. Finally, Section "Conclusion" summarizes the key characteristics of the proposed map and suggests directions for its practical implementation.

Related work

Several studies have been conducted to explore and improve the existing chaotic maps, focusing on enhancing their chaotic properties and evaluating their suitability for various applications.

Hua et al. (2019) introduced a sine chaotification model to enhance the chaotic range of one-dimensional chaotic maps using the sine function as a nonlinear chaotification transform. The model was evaluated for three chaotic maps: Logistic, Sine, and Tent maps. Performance evaluations show that the enhanced maps exhibit more intricate and complex behaviors than their original counterparts.

Hanis et al. (2019) proposed a modified Logistic map with enhanced key space and improved security. The proposed map demonstrates enhanced chaotic behavior and

a more comprehensive chaotic control parameter range from 3.9 to 6.27.

Pak et al. (2019) introduced novel enhancements to traditional 1D chaotic maps, including Logistic and Sine maps, derived from the output sequences of two identical 1D chaotic maps. The findings demonstrate an expanded range of control parameters compared to the original seed maps.

Herbadji et al. (2020) introduced an enhanced Quadratic map for application in an image encryption scheme. The proposed map exhibits superior performance characteristics compared to the classical Quadratic map, including better Lyapunov exponent and more extensive chaotic ranges.

Mansouri et al. (2020) introduced a novel 1D sine-powered chaotic system designed for image encryption applications. The results reveal robust chaotic behavior of the proposed map characterized by high sensitivity and randomness.

Khairullah et al. (2021) proposed two enhanced chaotic maps, the 1D improved Logistic map, and the 1D improved Quadratic map, to address the security shortcomings of traditional 1D chaotic maps in image encryption. These maps demonstrate superior performance metrics, including higher Lyapunov exponents, increased complexity, broader chaotic ranges, and greater sensitivity than existing 1D maps.

Moatsum Alawida (2024) introduced the enhanced Logistic map utilizing the anterior perturbation method. The findings report the proposed map to have enhanced randomness, improved data distribution, and reduced correlation. The study also introduces a novel pseudorandom number generator based on the combination of proposed maps, demonstrating its effectiveness in generating secure sequences.

Despite considerable efforts to enhance 1D chaotic maps, their chaotic behavior often remains constrained, limiting their effectiveness in providing robust security for cryptographic applications. This limitation is particularly evident in the case of the popular Logistic map, which, despite its simplicity and widespread adoption, requires improvements to expand its chaotic range and strengthen its security capabilities.

Proposed enhanced logistic map

The Logistic map is a one-dimensional discrete chaotic map. The iterative mathematical equation defines it:

$$x_{n+1} = r \cdot x_n \cdot (1 - x_n) \quad (1)$$

where ' r ' is the control parameter, and x_n and x_{n+1} represents the state of the system at time n and $n + 1$, respectively. Depending on the value of ' r ', the map exhibits various behaviors. These include stable fixed points,

periodic oscillations, and chaotic dynamics, with chaotic behavior typically emerging for ' r ' spanning between 3.57 and 4.

The enhanced Logistic map is a one-dimensional discrete chaotic map represented by the following iterative mathematical equation:

$$x_{n+1} = \text{mod} \left(\left(\frac{\pi}{x_n} \cdot r' \cdot (1 - x_n) \right), 1 \right) \quad (2)$$

where, x_n and x_{n+1} represents the state variable at time n and $n + 1$, respectively. The output lies in the range of 0 to 1. r' defines the control parameter defined in the interval $(0, +\infty)$.

The enhanced Logistic map modifies the traditional Logistic map by combining π , scaling, and the modulo operations. π is a fundamental mathematical constant that represents the ratio of a circle's circumference to its diameter. Irrespective of whether the circle is big or small, this ratio is always equal to π . This makes π a universal constant, crucial in various fields of science and mathematics (Berggren et al. 2004). In geometry, it is used for calculating properties of circles, spheres, and other curved shapes. Engineers use π in the design of circular structures and components, while astronomers rely on it for understanding planetary orbits and the universe's geometry. In computing, π is used for algorithm testing and random number generation. In physics, π appears in equations describing wave behavior, circular motion, and quantum mechanics. π being an irrational number cannot be expressed as a simple fraction. Its decimal expansion goes on infinitely without repeating or following any discernible pattern. This infinite and non-repeating decimal representation together with the randomness of digits make π an intriguing choice for increasing the complexity of chaotic maps as all its decimal places appear without any predictable sequence. In the enhanced Logistic map, π serves as a constant multiplier, introducing non-linear scaling effects that increase the complexity of the map's iterative behavior. Furthermore, the scaling operation by π/x_n makes the next map's state to be inversely proportional to the current state x_n . The value of x_n being small in the range of 0–1 makes the π/x_n term large, thus amplifying the impact of subsequent multiplication by the remaining term. Finally, the modulo operation by 1 ensures that x_{n+1} remains within the interval $[0, 1]$, creating a bounded system with interpretable and more manageable output suitable for applications where the variable must stay within specific bounds, such as in cryptography or signal processing.

Implementation

Methodology

The enhanced Logistic map has been evaluated in MATLAB R2021a on a 64-bit Windows 10 operating system running on an i7core processor with a clock frequency of 3.4 GHz and 8 GB of RAM.

Results

The evaluation of the proposed map has been conducted in terms of various chaos dynamical tests, including bifurcation diagram, Lyapunov exponent, 2D and 3D phase plots, cobweb plots, 0–1 test, approximate entropy, and sample entropy, and time sensitivity analysis.

Bifurcation diagram

It is a graphical representation used to illustrate the changes in the behavior of a chaotic map with variations in its control parameters (Muthu and Murali 2021a; Nazish and Banday 2023). It reveals critical information about the map's long-term behavior, delineates the parameter ranges for periodic and chaotic regions, and highlights bifurcation points where the map transitions from stable to chaotic behavior or undergoes periodic doubling. Figure 1 shows the bifurcation diagram of the

original Logistic map and its enhanced counterpart. For the seed Logistic map, a chaotic region exists for the control parameter ranging from $r = 3.57$ –4. Amidst this chaotic region, periodic or blank windows are observed. In contrast, the enhanced Logistic map exhibits a chaotic range extending indefinitely without any periodic windows in between. This implies that the proposed map exhibits highly complex and continuous chaotic behavior across a broader range of control parameter values compared to a classical Logistic map.

Lyapunov exponent

Lyapunov Exponent (Muthu and Murali 2021a; Nazish and Banday 2023) measures the separation rate of infinitesimally close trajectories in phase space, providing insight into the system's stability or chaos. Its value indicates whether the nearby trajectories diverge, suggesting chaotic behavior with a positive exponent, converge, indicating stability with a negative exponent, or remain unchanged, implying marginal stability with a zero exponent. Figure 2 shows the Lyapunov exponent observed for different control parameter values for the Logistic and proposed maps. The Lyapunov exponent of the Logistic map is positive only for the control parameter range of

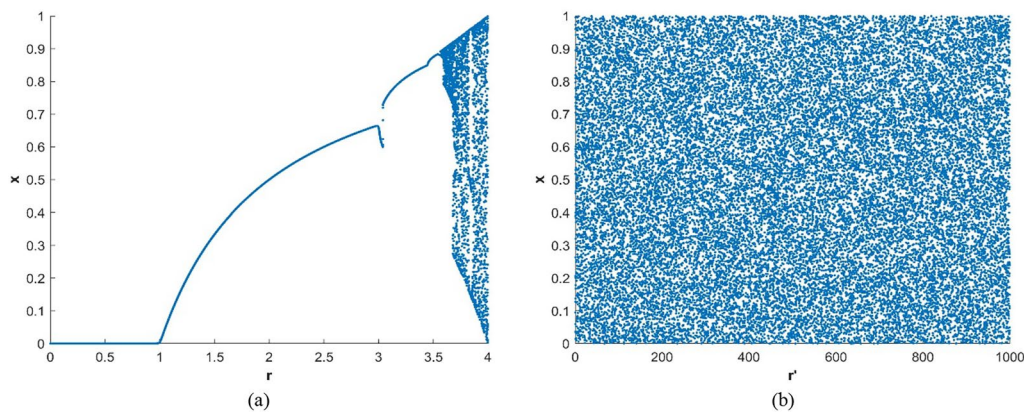


Fig. 1 Bifurcation diagram of the **a** original logistic map, **b** Enhanced Logistic Map

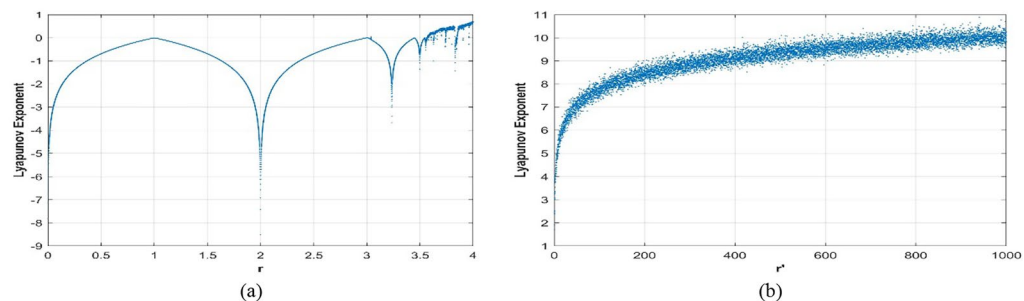


Fig. 2 Lyapunov exponent of the **a** original logistic map, **b** enhanced logistic map

3.57–4. Moreover, negative dips are observed even within this range, indicating the presence of periodic windows. In contrast, the Lyapunov exponent of the enhanced Logistic map remains positive throughout its entire control parameter range. Also, there are no intervals where the Lyapunov exponent turns negative, demonstrating persistent chaotic behavior without transitioning into periodic orbits.

Table 1 compares the proposed enhanced Logistic map with the seed and existing modified maps. Existing and seed maps typically feature narrow control parameter ranges, potentially compromising their security. Contrarily, the proposed enhanced Logistic map offers an infinitely expansive chaotic control parameter range with highly intricate non-linear dynamics, rendering it resilient to security attacks.

Moreover, the traditional and existing modified maps with narrow chaotic control parameter ranges often suffer from frail chaos, where the chaotic behavior is confined to limited and specific regions of the control parameter space. Even a slight modification to the control parameters in these maps can cause the system to

transition into non-chaotic behavior, severely limiting its usefulness in applications requiring consistent and robust chaos. Furthermore, the limited key space of these maps restricts their applicability in fields where security is paramount. Ideally, a key space of 2^{100} or higher is required to resist exhaustive key search attacks. While both the original and modified Logistic maps theoretically offer a key space of approximately 2^{106} , considering both the initial condition and control parameter, the practical key space of the original Logistic map is far less. This reduction occurs because the chaotic behavior of the original Logistic map is confined to a narrow range of control parameters (3.57–4), with even blank windows existing within this range. In contrast, the proposed enhanced Logistic map offers an infinite range of chaotic control parameters, eliminating these limitations. This infinite range ensures that the map maintains its chaotic properties across a much broader spectrum of control parameters, greatly expanding the practical key space and enhancing the security and robustness of the system in various applications.

Table 1 Comparison of proposed enhanced logistic map with seed and existing chaotic maps

Existing maps	Equation	Control parameter range
<i>Seed maps</i>		
Logistic Map (May 1976)	$x_{n+1} = r \cdot x_n \cdot (1 - x_n)$	$r \in [3.54, 4]$
Sine Map (Griffin 2013)	$x_{n+1} = \beta \cdot \sin(\pi \cdot x_n)$	$\beta \in [0.87, 1]$
Chebyshev Map (Liu et al. 2016)	$x_{n+1} = \cos(\gamma \cdot \cos^{-1}(x_n))$	$-1 \leq x_i \leq 1, \gamma \in [2, 6]$
Quadratic Map (Ramadan et al. 2016)	$x_{n+1} = \delta - x_n^2$	$\delta \in 1.5 \text{ to } 2$
Simple Quadratic Map (Muthu and Murali 2021b)	$x_{n+1} = \epsilon(x_n^2 - 5)$	$0.53 < \epsilon < 0.61$
Singer Map (Tubishat et al. 2022)	$x_{n+1} = \theta(7.86x_n - 23.31x_n^2 + 28.75x_n^3 - 13.302875x_n^4)$	$\theta \in 0.9 \text{ to } 1.08$
<i>Existing Modified Maps</i>		
Generalised Logistic Map (Krishnamoorthi and Murali 2017; Bhatnagar and Jonathan Wu 2012)	$f(x) = \frac{4\mu x(1-x)}{1+4(\mu^2-1)x(1-x)}$	$\mu \in [-0.6795, -0.4324]$
Enhanced Logistic Map (Hua et al. 2019)	$x_{n+1} = \sin(\pi \tilde{\alpha} x_n (1 - x_n))$	$\tilde{\alpha} \in (0, +\infty)$
Squared Sine Logistic Map (Carvalho and Leonel 2016)	$f(x) = rx(1-x) + \mu \sin^2(2\pi x)$	$r \in [3.5, 3.6]$
Logistic-Sine-Cosine (LSC) map (Hua et al. 2019)	$x_{n+1} = \cos(\pi(4rx_n(1-x_n) + (1-r)\sin(\pi x_n) - 0.5))$	$r \in [0, 1]$
Tent-logistic-cosine (TLC) map (Hua et al. 2019)	$x_{n+1} = \begin{cases} \cos(\pi(2rx_n \\ + 4(1-r)x_n(1-x_n) - 0.5)), & x_n < 0.5 \\ \cos(\pi(2r(1-x_n) \\ + 4(1-r)x_n(1-x_n) - 0.5)), & x_n \geq 0.5 \end{cases}$	$r \in [0, 1]$
Logistic-tent system (Zhou et al. Apr. 2014)	$x_{n+1} = \begin{cases} (rx_n(1-x_n) \\ + (4-r)x_n/2) \bmod 1 & x_i < 0.5 \\ (rx_n(1-x_n) \\ + (4-r)(1-x_n)/2) \bmod 1 & x_i \geq 0.5 \end{cases}$	$r \in [0, 1]$
Logistic-sine system (Zhou et al. 2014)	$x_{n+1} = (rx_n(1-x_n) + (4-r)\sin(\pi x_n)/4) \bmod 1$	$r \in [0, 1]$
Modified Logistic map (Hanis and Amutha 2019)	$z_{n+1} = \mu_z z_n(1-z_n)(1-z_n^2)$	$3.9 < \mu_z \leq 6.27$
Improved modified logistic (Attaullah and Jamal 2020)	$x_{n+1} = [\mu x_n(1-x_n) \times 10^j] - \mu x_n(1-x_n) \times 10^j$	$0 \leq \mu \leq 4$ $2 \leq j \leq 8$
<i>Proposed 1D Chaotic Map</i>		
Enhanced logistic map	$x_{n+1} = \text{mod}\left(\left(\frac{\pi}{x_n} \cdot r' \cdot (1-x_n)\right), 1\right)$	$r' \in (0, +\infty)$

2D and 3D phase plots

These graphical representations show a dynamical system's behavior over time. The 2D phase plot shows the relationship between two variables (x_i and x_{i+1}), whereas the 3D plot extends the concept to three variables (x_i , x_{i+1} , and x_{i+2}), in the state space.

Figure 3 shows 2D and 3D phase plots of enhanced Logistic map for control parameter value=100. It can be observed that the trajectories densely fill in the entire phase space with no discernable patterns.

Cobweb plot

These powerful visualization tools help understand the iterative dynamics of chaotic maps. These plots illustrate how the values evolve over iterations and how they converge or diverge based on the system's parameters. These reveal key characteristics such as fixed points, periodic orbits, and chaotic behavior. For a stable fixed point, the map converges to a point. The plot forms a closed loop for a periodic orbit, indicating repetitive behavior. For chaotic behavior, the plot shows a dense, irregular pattern. Figure 4 shows the cobweb plot of the original and enhanced Logistic maps for control parameter value=1. The green lines represent the iterative function of the maps used to generate the sequence. The blue lines denote the transitions between consecutive values. They are drawn to show how each value maps to the next iteration. The vertical blue lines represent the transition

from x_{i-1} to x_i , while the horizontal lines show the projection of the new value x_i onto the function curve. The plot demonstrates that, unlike the original Logistic map, which displays predictable and regular patterns, the enhanced Logistic map exhibits irregular rectangular patterns, indicating that the proposed map maintains chaotic behavior across the entire range.

0–1 Test

This test (Gottwald and Melbourne 2016) provides a simple yet effective method to determine whether a dynamical system is chaotic or non-chaotic. By analyzing the time series data of the proposed map, the test calculates the M–t, K–c, and p–q plots, as shown in Fig. 5. M–t plot calculates the mean square displacement 'M' of auxiliary variables and correlates it with the iteration index 't.' M(t) grows linearly with t for a chaotic system, resulting in test statistic 'K' close to 1, indicating exponential divergence of trajectories and sensitive dependence on initial conditions. In contrast, M(t) grows sublinearly or not at all for a non-chaotic system, yielding a test statistic 'K' close to 0, signifying predictable and stable behavior. For the enhanced Logistic map, the M–t plot exhibits a linearly increasing trend, affirming the chaotic nature of the improved Logistic map. The k–c plot measures the growth rate of the mean-square displacement. The 'k' values cluster around 1 for the proposed map, confirming solid

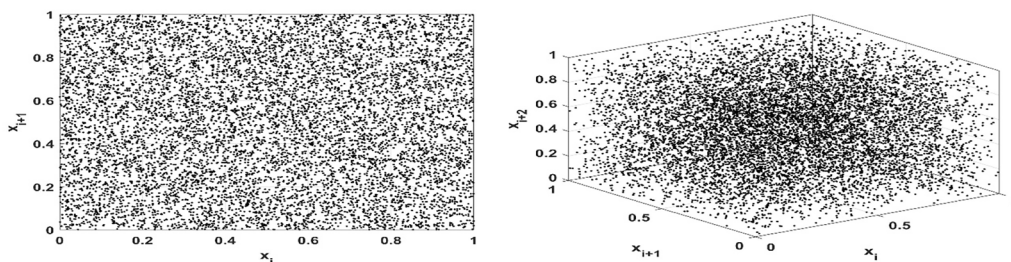


Fig. 3 a 2D and b 3D phase plots ($r' = 100$) of the enhanced logistic map

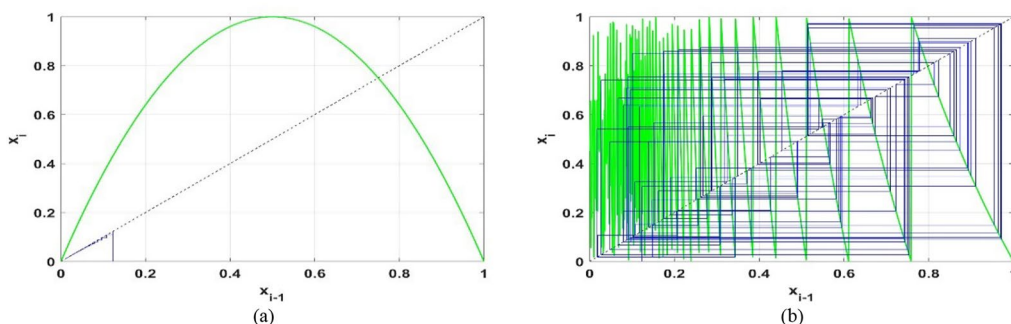


Fig. 4 Cobweb Plot ($r' = 1$) of the a Original logistic map b enhanced logistic map

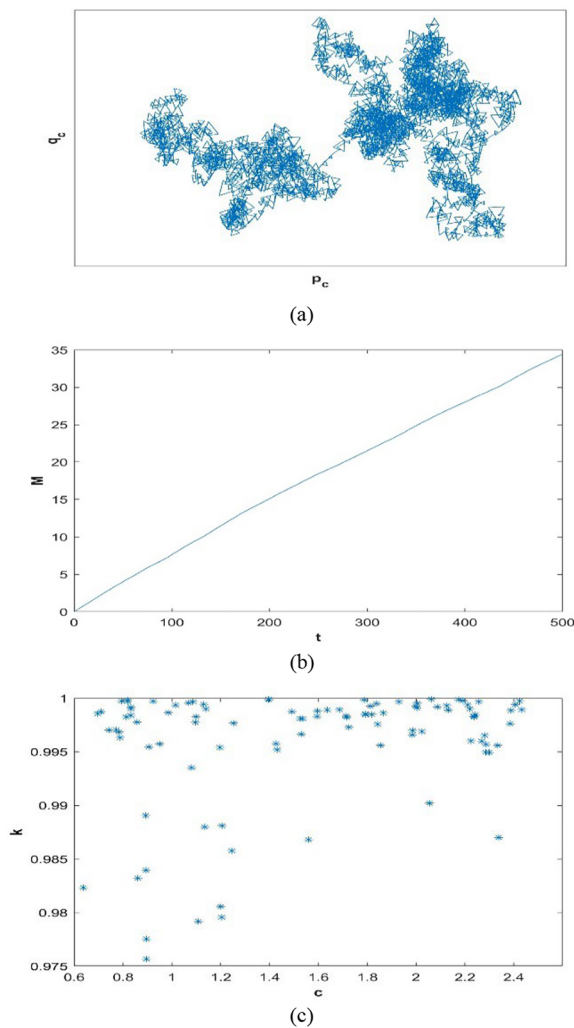


Fig. 5 0–1 Test plots: **a** p–q plot **b** M–t plot **c** k–c plot for the enhanced logistic map

Table 2 0–1 Test indicator value for enhanced logistic map

Chaotic map	Control parameter = 100
Enhanced logistic map	0.99834

chaotic behavior. The p–q plot visualizes the correlation between phase space projections. The plot for the enhanced map is irregular and complex, resembling Brownian motion, indicative of the system's intricate dynamics. Table 2 presents the indicator value for the improved map at a control parameter value of 100. This value is near the ideal 1, indicating that the proposed map exhibits robust chaotic dynamics.

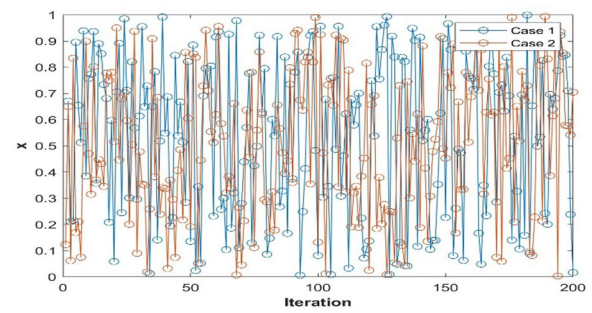


Fig. 6 Time sensitivity analysis for the enhanced logistic map

Table 3 Average, minimum, and maximum approximate entropy values for original logistic and enhanced logistic maps

Chaotic map	Average	Minimum	Maximum
Original logistic map	0.039961	−0.000153	0.64615
Enhanced logistic map	1.9818	0	2.0394

Time sensitivity

Time sensitivity analysis involves observing how minute variations in the initial conditions of a dynamical system affect its evolution over time. Figure 6 shows the time sensitivity analysis results performed for the enhanced Logistic map by varying the initial parameter slightly, from 0.12345 to 0.123451, while keeping the control parameter value constant at 100. Despite the minimal difference in the initial parameters, the observed results showed drastically different plots, highlighting the extreme sensitivity of the proposed map to the initial state.

Approximate entropy (ApEn)

ApEn (Pincus 1991) is a statistical measure used to quantify the amount of regularity and unpredictability in time series data. Lower values of ApEn indicate more regularity, while higher values suggest more complexity and randomness. Table 3 and Fig. 7 provide the ApEn values for the original and enhanced Logistic maps across varied control parameter values. The original Logistic map exhibits a relatively lower average ApEn value, suggesting a higher degree of regularity and predictability. In contrast, the enhanced map consistently exhibits higher ApEn values, which indicates a persistent state of greater complexity and randomness throughout its parameter space.

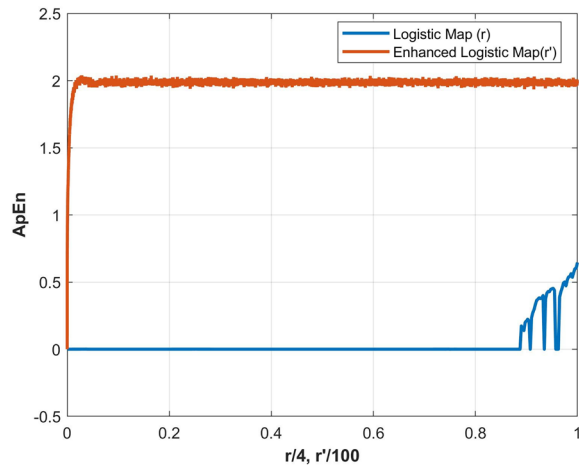


Fig. 7 Approximate entropy for the original and enhanced logistic maps

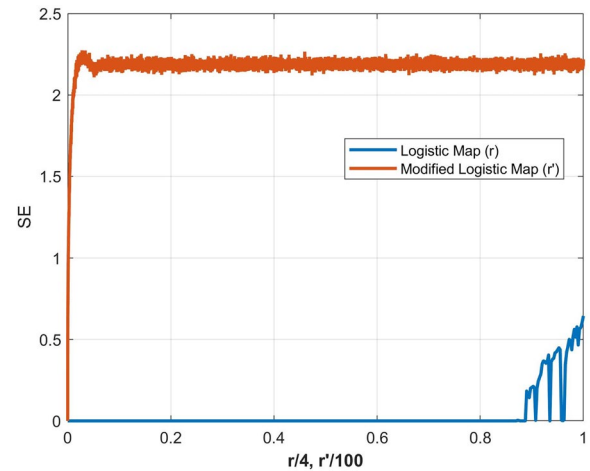


Fig. 8 Sample entropy values for the original logistic and enhanced logistic maps

Sample entropy (SE)

It is a statistical measure that assesses the complexity and irregularity in short, noisy time series data (Yentes et al. 2013). A higher SE value indicates greater complexity and unpredictability in the time series data, while a lower value suggests more regular and predictable behavior. Table 4 and Fig. 8 compare the Sample Entropy values for the seed Logistic map and its enhanced version. The seed Logistic map exhibits significantly lower average sample entropy values, indicating that it contains more repetitive and structured patterns. In contrast, the enhanced Logistic map has higher SE values across a wide range of control parameter values, suggesting that the enhanced map maintains persistently complex and irregular dynamics throughout its parameter space.

Proposed pseudo-random bit generator based on the enhanced logistic map

Pseudorandom bit generators (PRBGs) are integral to security applications, generating cryptographic keys, and initialization vectors. These generators produce bit sequences that appear random while being deterministic, generated from a seed value using mathematical algorithms. Chaotic map-based PRBGs utilize the deterministic and intricate behavior of chaotic maps to generate sequences of pseudorandom bit sequences. Their high sensitivity to initial seed values ensures vastly diverse

outputs, enhancing their ability to generate sequences that display randomness and possess statistical properties ideal for cryptographic applications.

Algorithm 1 presents the pseudocode for the pseudorandom bit generator (Fig. 9) utilizing the proposed enhanced Logistic map to produce a long pseudo-random bitstream, with detailed steps provided below:

Step 1 Initialize the PRBG with an initial seed value ' x ' in the range $[0, 1]$. This seed value which represents the initial state of the enhanced Logistic map, serves as the starting point for the iterative generation of pseudo-random bits.

Step 2 Set the control parameter R for the enhanced Logistic map to a specific value (e.g., $R = 100$) within the range $(0, +\infty)$.

Step 3 Define constants such as PI (π) and specify the desired length of the bit sequence, denoted as *bitlength*.

Step 4 Predefine an empty vector '*bits*' initialized with NaN values and a length equal to '*bitlength*' (as shown in equation (III)) to store the generated bits.

$$\text{bits} = \text{NaN} * \text{zeros}(1, \text{bitlength}) \quad (3)$$

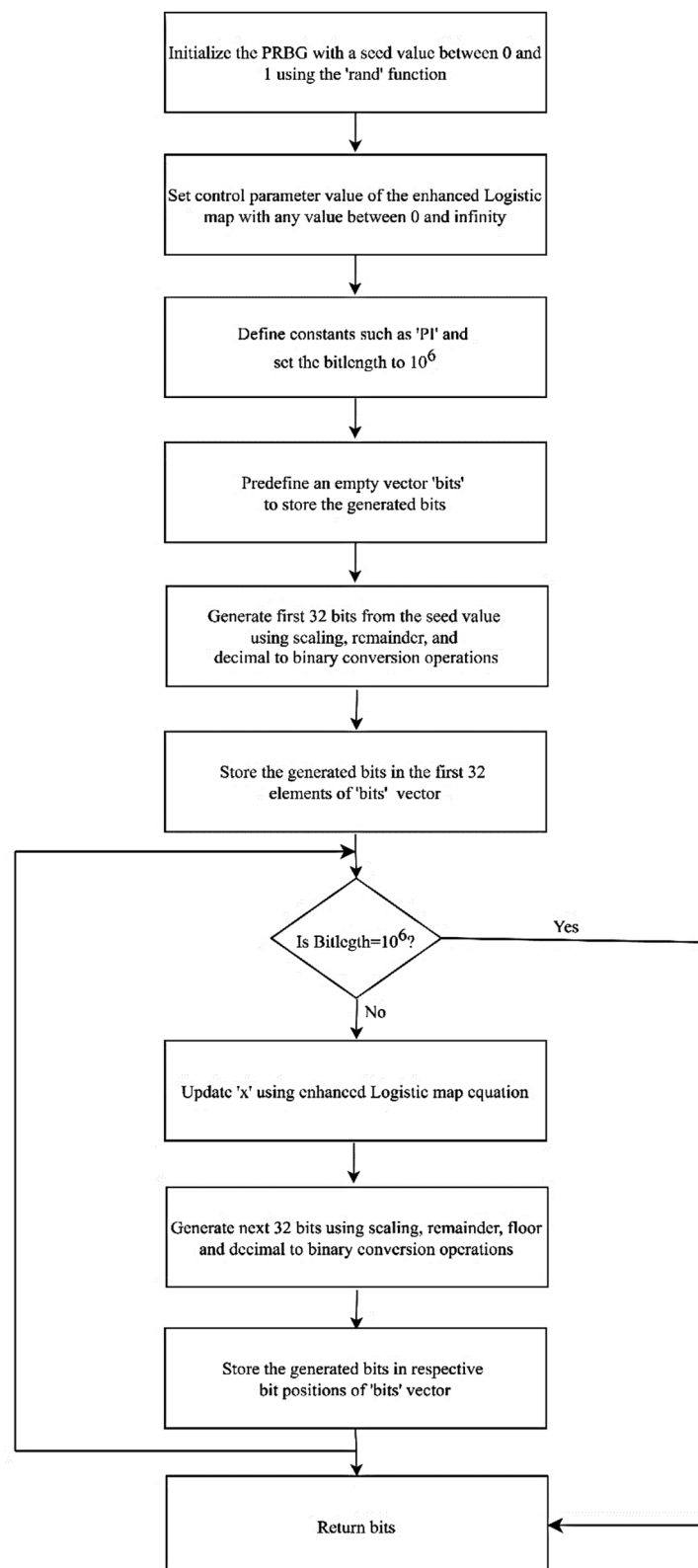
Step 5 Generate the first 32 bits using the initial value of x . This involves scaling x , computing the remainder, flooring the result, and converting it into a binary representation using the *de2bi* function. Store the resulting binary vector in the first 32 elements of the '*bits*' vector.

$$\text{bits}(1 : 32) = \text{de2bi}\left(\text{floor}\left(\text{rem}\left(10^{12} * x, 2^{32}\right)\right), 32\right) \quad (4)$$

The '*rem*' function in the formula (4) calculates the remainder when $10^{12} * x$ is divided by 2^{32} . This function is crucial for ensuring that the resulting value fits within the 32-bit boundary which is essential for generating 32-bit blocks of the bit stream.

Table 4 Average, minimum, and maximum sample entropy values for original logistic and enhanced logistic maps

Chaotic map	Average	Minimum	Maximum
Original logistic map	0.038655	0	0.64453
Enhanced logistic map	2.1808	0	2.2742

**Fig. 9** Proposed enhanced logistic map-based PRBG

Step 6 Iteratively update x using the enhanced Logistic map formula as defined in Eq. (2) to generate subsequent bits until the desired bit length is achieved. Begin iterating from the second block of 32 bits up to $\lceil \text{bitlength}/32 \rceil$ blocks. In each iteration, x is updated using the enhanced Logistic map formula $x = \text{mod}((\pi * R * (1 - x))/x, 1.0)$. New 32 bit blocks are generated by scaling x , computing the remainder, flooring, and converting to binary, which is then stored in the respective positions in the bits vector.

The proposed PRBG effectively generates a pseudorandom bitstream of length “*bitlength*”, leveraging the chaotic properties of the enhanced Logistic map initialized with x and controlled by π and R .

Algorithm 1 GenerateRandomBitstream(x, R)

Input: Initial parameter x (random number between 0 and 1), R (parameter)

Output: Array of bits, *bits*

$\pi \leftarrow 3.14159265358979323846$

bitlength $\leftarrow 1000000$

bits $\leftarrow$ array of NaN values of length *bitlength*

// Generate first 32 bits

bits[1 : 32] $\leftarrow \text{de2bi}(\lfloor \text{rem}(10^{12} \cdot x, 2^{32}) \rfloor, 32)$

for $i \leftarrow 2$ to $\lceil \frac{\text{bitlength}}{32} \rceil$ do

 // Update x using the given formula

$x \leftarrow \text{mod}((\pi \cdot R \cdot (1 - x))/x, 1.0)$

 // Generate next 32 bits and store them in the bit vector

bits[($i - 1$) $\times$ 32 + 1 : $i \times$ 32] $\leftarrow \text{de2bi}(\lfloor \text{rem}(10^{12} \cdot x, 2^{32}) \rfloor, 32)$

end for

return *bits*

▷ Define π

▷ Length of the bitstream

▷ Predefine an empty vector for the bits

▷ Loop to generate the rest of the bits

Table 5 lists the operations involved in generating each 32-bit block during a single iteration of the pseudo-random bitstream until the desired length is achieved. The constant values 10^{12} , 2^{32} , and π can be precomputed and are not counted as separate operations. With a total of 9 operations per block of 32 bits, the algorithm averages approximately 0.28125 operations per bit, underscoring its computational efficiency in generating a large sequence of high-quality pseudorandom bits. Furthermore, the entire pseudo-random bit-sequence generation takes only 2.1208 seconds for generating 10^6 bits, demonstrating the PRBG’s efficiency and suitability for high-speed cryptographic applications.

NIST statistical test for PRBG generated bit-sequence

The NIST test suite (Bassham et al. 2010) encompasses fifteen tests designed to assess the randomness of sequences generated by random or pseudorandom number generators. The binary sequence under test qualifies for a specific test if the computed p-value is more significant than the pre-determined threshold, typically 0.01. The enhanced Logistic map-based PRBG-generated bit

sequence is subjected to these comprehensive tests. The results in Table 6 demonstrate that the sequence exhibits all characteristics of randomness, uniformity, and independence required for secure cryptographic applications.

The overall results demonstrate exceptional performance of the enhanced Logistic map across a comprehensive suite of chaotic dynamical tests, consistently exhibiting characteristics of robust chaotic behavior. The bifurcation diagram reveals an infinite chaotic range, distinct from the classical Logistic map, with no periodic windows interrupting the chaos. This is further validated by the Lyapunov exponent analysis, which shows consistently positive values across the entire control parameter range. The 2D and 3D phase plots exhibit complex, non-repeating trajectories typical of chaotic systems. The cobweb plot reveals intricate, rectangular patterns characteristic of chaos. At the same time, the 0–1 test result, with $M(t)$ growing linearly and K values clustering around 1, unequivocally affirms the chaotic nature of the map. Moreover, the high approximate entropy and sample entropy values consistently observed across all control parameter

Table 5 Number of operations required for a single iteration of the enhanced logistic map-based PRBG

Operation	Count
Multiplication	3
Division	1
Modulus	1
Subtraction	1
Decimal to binary conversion	1
Floor	1
Remainder	1
Total operations	9
Operation/Bit	0.28125

ranges reflect the map's intricate complexity and unpredictability. Time sensitivity analysis reveals the map's extremely sensitive dependence on initial conditions. Thus, the enhanced Logistic map excels in all evaluations, demonstrating unparalleled chaotic behavior. Furthermore, the PRBG-generated bit sequence generates a 32-bit sequence in a single iteration, and successfully qualifies all NIST tests, confirming its high-speed and strong randomness characteristics. Thus, the proposed map and PRBG offer highly reliable and robust solutions, ideal for securing resource-limited applications requiring high complexity and unpredictability.

Table 6 NIST evaluation results for the enhanced logistic map-based PRBG generated bit sequence

NIST test	Value	Status
Monobit frequency	0.618484042	Passed
Block frequency	0.049756895	Passed
Runs	0.470773974	Passed
Longest runs of ones	0.791859411	Passed
Binary matrix rank	0.185901474	Passed
Discrete fourier transform	0.330390049	Passed
Non-overlapping template matching	0.766501877	Passed
Overlapping template matching	0.3643286	Passed
Maurer's universal statistical	0.218117333	Passed
Linear complexity	0.025358363	Passed
Serial	0.913995697	Passed
Approximate entropy	0.630093451	Passed
Cumulative sums forward	0.754256444	Passed
Cumulative sums reverse	0.345049598	Passed
Random excursions	0.388632592	Passed
Random excursions variant	0.022866558	Passed
Success count	15/15	

Conclusion

The paper proposes an enhanced Logistic map with highly intricate chaotic behavior featuring an extended infinite chaotic control parameter range. This enhancement is validated through chaos dynamical tests, including bifurcation diagram, Lyapunov exponent, 2D and 3D phase plots, cobweb plots, 0–1 test, approximate entropy, sample entropy, and time sensitivity analysis. Furthermore, the enhanced Logistic map-based Pseudo-Random Bit Generator (PRBG) demonstrates exceptional performance in generating high-quality pseudo-random bits. Its lightweight implementation ensures minimal computational overhead while maintaining robust security features. With a remarkably low average of approximately 0.28125 operations per bit and the ability to produce 32 bits in a single iteration, the PRBG achieves notable speed and efficiency. Additionally, the PRBG-generated bit sequence successfully passes all fifteen NIST statistical tests for randomness, underscoring its suitability for high-security applications such as pseudorandom number generation, secure data transmission, and encryption. A promising step forward is to validate the practical applicability and efficacy of the proposed map, specifically focusing on its utilization in lightweight pseudorandom bit generation and cryptographic primitives.

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Author contributions

Mir Nazish- Conceptualization, Supervision, Methodology, Investigation, Writing, original draft preparation, Editing, Reviewing, Experimentation, Prepared Figures and Tables. Munika Javid—Methodology, Conceptualization, Writing. M. Tariq Bandy- Supervision, Conceptualization, Methodology, Reviewing, Revision, Editing, Writing.

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Availability of data and materials

The data supporting this study's findings are available from the corresponding author upon reasonable request.

Declarations

Competing interests

The Authors declare that they do not have any conflict of interest.

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